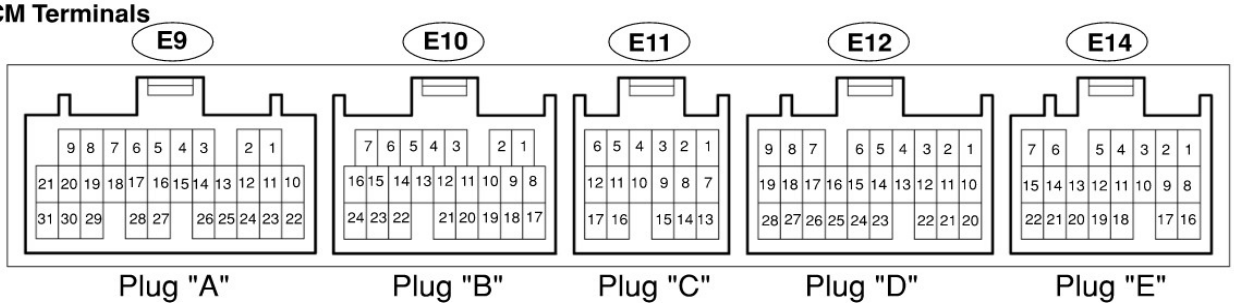
	Year
4Runner:	1999-2000
Tacoma:	2000-2001
Engine:	5VZ-FE (3.4L V6)
Board:	uaEFI Adapter Board V1.1

Notes:
California or Federal spec emissions (one or two catalytic converters from factory)
2000: California-style emissions (two catalytic converters from factory and upstream Air Fuel Ratio sensor)

****WARNING**** Do not plug your ECU into your laptop/computer if your laptop/computer is plugged into the vehicle's 12v charging port!

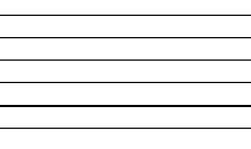
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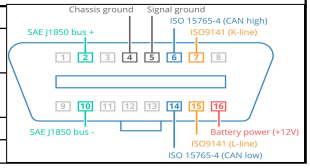


OPEN PINS

Note: Populate open slots in ECU connectors with MLC040III-SKT1 (OEM p/n 82998-12690) or MLC040III-SKT2 (OEM p/n 82998-12740) pins

Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E9	A	5	Lowside Output (with PWM)	B17 Low Side Output 3	PWM Ground Output. Has flyback diode protection.
E9	A	17	Ignition Output 4	B11 Coil 4	**MUST be used with an igniter or smart coils**
E9	A	18	Ignition Output 5	B13 Coil 5	**MUST be used with an igniter or smart coils**
E9	A	19	Ignition Output 6	B10 Coil 6	**MUST be used with an igniter or smart coils**
E9	A	23	G-	N/A	For separated negative VR leads in the factory wiring harness. Connect camshaft VR- here.
E9	A	24	+B	N/A	+12V battery voltage
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
E10	B	1	HB1+		H-Bridge 1 +
E10	B	3	HB1-		H-Bridge 1 -
E10	B	11	PPS2	C4 PPS2	Pedal Position Sensor 2 (Can be used for any 0-5v Analog Input)
E10	B	13	TPS2	C14 TPS2	Throttle Position Sensor 2 (Can be used for any 0-5v Analog Input)
E10	B	20	PPS1	D6 PPS1	Pedal Position Sensor 1 (Can be used for any 0-5v Analog Input)
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
WARNING If your vehicle has a factory 17-pin center connector, do **NOT** plug it into this adapter board! Leave it disconnected! It is for the immobilizer system and is no longer used Use a MLC040IIIHD-17S (OEM p/n 353027-1) 17-pin connector and populate with MLC040III-SKT1 (OEM p/n 82998-12690) or MLC040III-SKT2 (OEM p/n 316838-1) pins					
E11	C	1	IP	N/A	For Bosch 17025 (LSU 4.9) WBO (Connector Pin 1)
E11	C	2	H-	N/A	For Bosch 17025 (LSU 4.9) WBO Heater - (Connector Pin 3)
E11	C	3	CalR	N/A	For Bosch 17025 (LSU 4.9) WBO (Connector Pin 5)
E11	C	4	7th Injector	B9 MR Weak LS 1	Connect 7th injector negative here, and positive to +B (12V battery)
E11	C	5	MAP	D9 MAP	Manifold Absolute Pressure Sensor (Can be used for any 0-5v Analog Input)
E11	C	6	AUX2	C3 AUX2	Analog Input #2 (0-5v)
E11	C	7	VS/IP	N/A	For Bosch 17025 (LSU 4.9) WBO (Connector Pin 2)
E11	C	8	H+	N/A	For Bosch 17025 (LSU 4.9) WBO Heater + (Connector Pin 4)
E11	C	9	VS	N/A	For Bosch 17025 (LSU 4.9) WBO (Connector Pin 6)
E11	C	10	5V Sensor	N/A	+5V for sensors
E11	C	11	5V Sensor	N/A	+5V for sensors
E11	C	12	Flex	D5 FLEX	Flex Fuel Signal
E11	C	13	+B	N/A	+12V battery voltage
E11	C	14	+B	N/A	+12V battery voltage
E11	C	15	AUX1	D1 AUX1	Analog Input #1 (0-5v)
E11	C	16	Sensor Ground	N/A	For sensor grounds only
E11	C	17	Sensor Ground	N/A	For sensor grounds only
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
E12	D	1	Digital Input 2	MEGA_EPIC_D30	Ground activated
E12	D	5	Digital Input 1	MEGA_EPIC_D31	Ground activated
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
E14	E	18	CAN1H	N/A	CANBUS High. If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #6
E14	E	19	CAN1L	N/A	CANBUS Low. If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #14
E14	E	20	Lowside 1	MEGA_EPIC_D45	On/Off Ground Output
E14	E	21	Lowside 2	MEGA_EPIC_D46	On/Off Ground Output

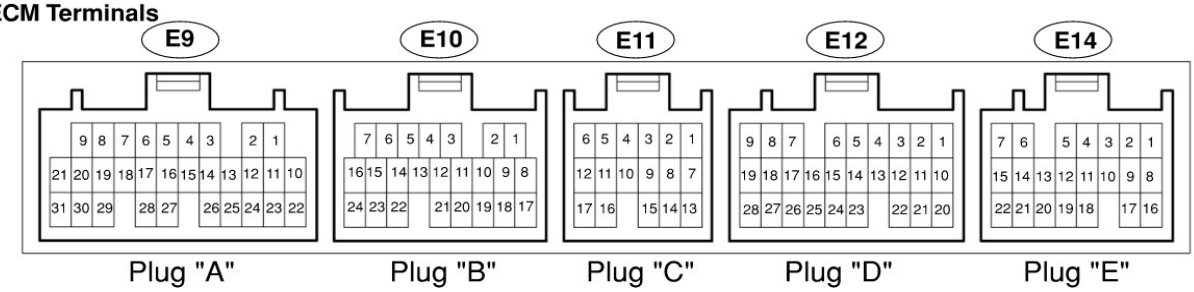




Year	Notes:
4Runner: 1999-2000	California or Federal spec emissions (one or two catalytic converters from factory)
Tacoma: 2000-2001	2000: California-style emissions (two catalytic converters from factory and upstream Air Fuel Ratio sensor)
Engine: 5VZ-FE (3.4L V6)	
Board: uaEFI Adapter Board V1.1	

****WARNING**** Do not plug your ECU into your laptop/computer if your laptop/computer is plugged into the vehicle's 12v charging port!

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POPULATED PINS

Note: Populate open slots in ECU connectors with MLC040III-SKT1 (OEM p/n 82998-12690) or MLC040III-SKT2 (OEM p/n 82998-12740) pins

Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E9	A	1	#30	B4 Injector Output 3	Injector #3
E9	A	2	#40	B3 Injector Output 4	Injector #4
E9	A	3	#50	B2 Injector Output 2	Injector #5
E9	A	4	#60	B1 Injector Output 6	Injector #6
E9	A	7	S1	MEGA_EPIC_D39	Shift Solenoid 1
E9	A	8	S2	MEGA_EPIC_D41	Shift Solenoid 2
E9	A	9	SL	MEGA_EPIC_D5	TCC Lockup Solenoid
E9	A	10	G2	C7 HALL3	Cam Signal
E9	A	11	IGT1	B15 Coil 1	Ignition Coil #1
E9	A	12	IGT2	B14 Coil 2	Ignition Coil #2
E9	A	13	IGT3	B12 Coil 3	Ignition Coil #3
E9	A	14	SP2+	C18 VR1+ Discrete	Speed Sensor 2
E9	A	15	RSC	B18 Low Side Output 2	Idle Air Control Valve - Close
E9	A	16	RSO	B7 Low Side Output 1	Idle Air Control Valve - Open
E9	A	22	PWR	MEGA_EPIC_D24	ECT PWR Button (Automatic Transmission)
E9	A	27	KNK1	D14 Knock Input	Knock Sensor Bank 1 (use selector switch on carrier board to choose KNK1 or KNK2)
E9	A	28	KNK2	D14 Knock Input	Knock Sensor Bank 2 (use selector switch on carrier board to choose KNK1 or KNK2)
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
E10	B	4	HTAF	MEGA_EPIC_D13	Upstream O2 Sensor Heater
E10	B	5	#10	B6 Injector Output 1	Injector #1
E10	B	6	#20	B5 Injector Output 2	Injector #2
E10	B	7	EVP1	B8 LS2_NO_FLYBACK	EVAP Solenoid
E10	B	9	PSW	MEGA_EPIC_D27	Power Steering Switch
E10	B	10	VG	C15 AUX3	Mass Air Flow Sensor Signal
E10	B	12	AF1+	MEGA_EPIC_ADC_0	Upstream Air Fuel Ratio Sensor Signal
E10	B	12	OX1	MEGA_EPIC_ADC_1	Upstream Narrowband Oxygen Sensor Signal
E10	B	14	THW	D16 CLT Coolant	Engine Coolant Temperature
E10	B	15	OIL	MEGA_EPIC_ADC_12	Automatic Transmission Fluid Temperature
E10	B	16	NE+	C6 HALL2	Crank Sensor
E10	B	22	THA	D15 IAT	Intake Air Temperature
E10	B	23	VTA (TPS)	D13 TPS1	Throttle Position Sensor
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
E12	D	2	R	MEGA_EPIC_D34	Reverse (Automatic Transmission)
E12	D	3	2	MEGA_EPIC_D35	2nd Gear Selector (Automatic Transmission)
E12	D	4	IDLO	MEGA_EPIC_D40	Idle Sensor
E12	D	6	ODMS	C9 BUTTON 3	Overdrive Lockout Switch
E12	D	7	OILW	MEGA_EPIC_D42	A/T Fluid Overheat Light
E12	D	8	OXS	MEGA_EPIC_ADC_15	Downstream Oxygen Sensor
E12	D	9	HTS	MEGA_EPIC_D12	Downstream Oxygen Sensor Heater
E12	D	10	OD2/ODLP	C9 BUTTON 3	Overdrive Lockout Switch / Light
E12	D	11	PWR	MEGA_EPIC_D24	ECT PWR Button (Automatic Transmission)
E12	D	12	L	MEGA_EPIC_D33	Low Gear Selector (Automatic Transmission)
E12	D	13	ACT	MEGA_EPIC_D49	A/C Terminate
E12	D	18	4WD/AWD	MEGA_EPIC_D29	4WD Indicator
E12	D	19	L4	MEGA_EPIC_D25	Transfer Case Low Range
E12	D	20	NSW	MEGA_EPIC_D32	Neutral Safety Switch
E12	D	22	SP1	C5 HALL1	Speed Sensor 1
E12	D	25	AC1	MEGA_EPIC_D23	A/C Request
Connector	**Plug**	**Pin**	**Function**	**Tuner Studio Channel Name**	**Notes**
E14	E	3	FC	B16 Low Side Output 4	Fuel Pump
E14	E	6	W	MEGA_EPIC_D43	Check Engine Light
E14	E	7	STA	MEGA_EPIC_D28	Cranking Signal
E14	E	15	STP	MEGA_EPIC_D22	Brake Light
E14	E	17	PTNK	MEGA_EPIC_ADC_14	EVAP Pressure